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Summary

Senior Data & Applied Scientist and org-recognized AI Champion (9+ years) shipping production AI and ML at cloud scale—autonomous/multi-agent LLM systems, behavioral embeddings, experimentation programs, and revenue-critical data platforms; recent impact includes ~\$45M/month influenced revenue recovered and multi-hour incident triage cut to minutes.

Experience

Microsoft, Redmond, WA | *Sr. Data & Applied Scientist*

Nov 2020 – Current

AI & Agentic Systems:

- Cut customer-health time-to-insight from **days to <2 minutes** by architecting an enterprise LLM agent for **Mission-Critical Customers (MCC)** that turns natural-language questions into governed **live Kusto/ADX queries** and returns PM-ready narratives, grounded on a curated catalog of 13 metric domains.
- Reduced multi-hour on-call incident investigations to **minutes** with an autonomous, **multi-agent incident-investigation agent**—a planner/executor LLM loop on Azure OpenAI with context compaction, confidence scoring, and run tracing (FastAPI on Azure); shipped two reusable **Model Context Protocol (MCP)** tool servers (one almost entirely AI-generated) serving governed query and telemetry tools to Copilot, VS Code, and internal agents.

Applied ML & Data-Quality Platform:

- Enabled fleet-scale similarity search, segmentation, and embedding-drift anomaly detection across **millions of accounts** by designing an **account-behavior embedding model** (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.
- Caught silent upstream data regressions—including an **~88% row-collapse**—before they reached dashboards or Finance by building a self-service **data-quality monitoring service** (Python/FastAPI, anomaly detection, 76 tests, Azure Container Apps) adopted across teams, plus an hourly table-health profiling layer over **47 datasets**.
- Built and operated **large-scale telemetry pipelines and governed metrics**, applying statistical analysis and anomaly detection to validate data quality and turn ambiguous signals into decision-ready insights.

Partner Revenue & Business Reporting:

- Recovered and sustained **~\$45M/month in influenced partner revenue** (<1% variance vs Finance, ~4× partner coverage) by rebuilding a 20+ component attribution pipeline with **multi-partner credit** replacing last-writer-wins—lifting partner-attributed influenced revenue **>10% month-over-month**.
- Collapsed monthly business-review reporting from a multi-day manual process to a governed **~5-minute pipeline** (>95% effort reduction), now in regular leadership use; shipped Files/Disk/Blob business-insights dashboards on a curated dataset layer.

Reliability & Experimentation:

- Built an adaptive pre-release **stress-testing program** for Azure Storage informing per-build ship/hold decisions; helped prevent **10+ Sev1/Sev2 customer incidents** by catching top-severity regressions before rollout.
- Ran **thousands of controlled simulations per build** across 20 workload types, distilling ~500 failure signals into **15 recurring failure modes** and confirming 50+ issues with reproduction steps and quantified impact.
- Prioritized high-yield test scenarios with **online learning (multi-armed bandits)** and tuned scoring thresholds on outcomes, cutting false positives **60%**; led a 2-person effort and authored the design doc + internal white paper.

Mission-Critical Customer Health:

- Drove the **Mission-Critical Customers (MCC) health program** and a unified customer-health view across **8+ engineering/support teams**, cutting executive/customer readiness insight from **3–5 days to near real-time**—with ~50% fewer >10-day aging issues and **~16% faster time-to-mitigate**.
- Helped reverse churn/downtrend risk for **10+ critical customers**—each representing multi-millions in monthly Azure Storage revenue—by accelerating early detection and action on customer-impacting issues.
- Shipped **15+ Azure Advisor recommendations** adopted across hundreds of thousands of customer accounts to improve cost and performance, partnering with engineering and support to validate and operationalize.

Leadership & Recognition:

- As an org-recognized **AI Champion**, drove AI adoption across the team—co-led a 90-minute internal AI Day-of-Learning, mentored engineers (reviewed 17 PRs, onboarded a new hire), and led cross-org incident response (4-team Sev-2).

Walmart Labs, Bentonville, AR | *Senior Data Scientist*

Feb 2019 – Nov 2020

- Delivered **~7% category revenue lift** across 50 stores with an end-to-end **planogram optimization** system, cutting planning cycles from weeks to hours.
- Built **product association + substitution analytics** (behavior tree + demand-transfer “universes”) to support assortment and replenishment decisions.
- Led **4 DS/analysts + 2 interns** to productionize a repeatable item-level placement workflow; earned a Spot Award.

Integral Ad Science, New York, NY | *Data Scientist Intern*

June 2018 – Aug 2018

- Built a PySpark contextual text-classification pipeline processing **~1M URLs/day across 846 categories** (88% accuracy); improved label quality via correlation analysis.

Mu Sigma Business Solutions, Bengaluru, India | *Data Scientist*

Oct 2015 – July 2017

- Estimated **customer lifetime value (CLTV)** with probabilistic/regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

Education

University of Illinois at Chicago (UIC), Chicago, IL | *MS-Business Analytics*

Aug 2017 – Dec 2018

SGSITS, Indore, India | *B.E. in Computer Engineering*

Aug 2011 – July 2015

Skills

- **AI / ML:** Generative & agentic AI, autonomous/multi-agent systems & agent orchestration (planner/executor, tool-calling, RAG/grounding), Model Context Protocol (MCP) servers & clients, Azure OpenAI, LLM observability/tracing, LLM/agent evaluation, vector embeddings & autoencoders, FAISS, anomaly detection, multi-armed bandits, classification (NLP), production ML
- **Languages & Tools:** Python, SQL, Azure Data Explorer (Kusto), PySpark, R, FastAPI, Power BI, Tableau, Azure DevOps, Git
- **Platforms & Functional Areas:** Azure (App Service, Container Apps, Data Factory), large-scale telemetry & data-quality pipelines, experimentation & causal measurement, partner/revenue attribution & business reporting, customer & retail analytics, contextual classification at scale